INSTITUTO POLITÉCNICO NACIONAL
ESCUELA SUPERIOR DE CÓMPUTO

Introduction to Cryptography

Lab 05: Bit permutation *March 19,2026*

1. Warm-up exercises

Do the following exercises by your own in C/C++ programming language. Complete all the exercises in this section and upload screenshots of your source code, screenshots of your program running showing the results obtained and the calculations you did by hand to Microsoft Teams, before the session ends. Each exercise done in this section has a value of 0.5 points.

1. Declare a variable `data` of type `unsigned char` and do the following:
   - Asign to `data`, any printable ASCII character.
   - Use only `printf` to print the value of data as a character, as an hexadecimal value and as an integer.
2. Using the same variable `data`, use the bitwise operator << to do a left shift of only one bit, print the value of `data` as a character, as an hexadecimal, as an integer. Repeat this process eight times. What can you conclude by observing the results?
3. Initialize `data`, with letter 'A' use the bitwise operator >> to do a right shift of only one bit, print the value of `data` as a character, as an hexadecimal, as an integer. Repeat this process eight times. What can you conclude by observing the results?
4. Declare two variables `v1`, `v2` both of type `unsigned char`.Asign to `v1`, `v2` any value , then use the bitwise operators AND (&), OR (|) and XOR ($\wedge$), between them. Print the result as an hexadecimal and as integer. Do each bitwise operations by hand with the given values, and compare the results that you obtained with those given by your program.
5. Declare a variable `data2`, asign it any value. Do a bit mask using the bitwise operator AND (&) between `data2` and another convenient value to obtain the most significant bit of `data2`. Do this operation by hand and compare your result with the result of your program.
6. Use again variable `data2`, asign it any value. Do a bit mask using the bitwise operator AND (&) between `data2` and another convenient value to obtain the fourth least significant bits of `data2`. Do this operation by hand and compare your result with the result of your program.

2. Programming exercises

Please do the following exercises by your own using C/C++ programming language and bitwise operators.

1. Write a program to get the nth bit of a number. Explain using an example how it works.
2. Write a program to set nth bit of a number. Explain using an example how it works.
3. Write a program to count leading zeros in a binary number. Explain using an example how it works.

3. Products

Write a document to present your results. Your document must include

1. Personal information, date of the lab session and the topic that we are studying in this lab session.
2. Source code for Section 2 and your explanation with examples for each exercises.